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Introduction

Unlike other forms of forensic evidence, audio and video recordings can provide a real-time, eyewitness account a crime so investigators can watch or hear what transpired. For instance, a surveillance video captures a bank robbery in progress, or a hidden camera records an undercover sting operation.



INTRODUCTION

CCTV footage, video and audio recorders can be found in

- Business,
- Traffic intersections
- Parking lots
- Bank machines
- Police vehicle dashboards
- Cell phones etc

Repair and Recovery Evidence

- **Analog/digital magnetic tape-** It may need to be spliced back together
- **Digital World (CDs, DVDs, Cell phones, portable cameras etc)-** evidence can be damaged by heat, misuse, environmental conditions of a crime scene or simply on purpose by an offender.

Authentication of recordings

- If the ambient sound present on a recording changes abruptly, this could indicate where the recording took place suddenly changed.
- The volume and tone of a voice on the recording can provide clues as to distance and spatial relationships within a scene.
- Lighting conditions can be examined to estimate the time of day or environmental conditions at the same time of the recording.
- Unnatural waveform present in the audio or video signal may indicate that an edit has been made.
- A physical identifier may be present in the signal on magnetic tape that can identify it as a copy or indicate that it was recorded on a particular device.

Types of video and audio

- Digital video and audio
- Analog video and audio

Digital video and audio

- Half of all closed-circuit television evidence seized by police today is digital and file-based
- Some are in Magnetic Tapes
- Digital video recorders come in two general types:
  - embedded stand-alone
  - PC-based
- Both types generally record the audio and video to **hard drives, secure digital(SD) cards.**

Digital video and audio recording devices

- CD/DVD writing capabilities,
- USB for output
- some, although digital, may only have analog outputs

Analog video and audio

- **Every playback of the tape will degrade the recorded images.**
- Prior to ejecting the tape, the investigator will make sure the
  - tape is stopped
  - document everything on the display,
  - then eject the tape and
  - remove the write protection tab to prevent it from being recorded over.
  - A copy of the tape should then be made for all future viewing, preserving the original video evidence.

Important evidence to be recorded

- **make and model of the recording device**
- **how many cameras are connected and recording**
- **whether it is recording in time-lapse mode**
- **the current time/date, and the time/date on the recorder's display**
- sketch the **cameras' positions.**

Who Conducts the Analysis

American Society of Crime Laboratory Directors Laboratory Accreditation Board differentiates the audio and/or video analysis into four parts

- forensic audio analysis
- forensic video analysis
- image analysis
- computer forensics

Certification Programs

Two **certification programs** currently exist in the discipline of **Video Analysis:**

- Law Enforcement and Emergency Services Video Association
- International Association for Identification

In the field of forensic audio, training courses are available, but no specific certification currently exists



How the analysis is performed

- First step is to simply listen to or view the recorded footage
- Locate the area of interest to be enhanced or examined using specialized devices and software
- Before processing a working copy of the evidence may be created.

Video Enhancement Technique

A variety of enhancement techniques can be employed on video evidence. It is important that the best video recording be submitted to obtain the best enhancement results. **Limitations on the enhancement process** may exist if an analog copy or digital file that has **undergone additional compression** is submitted for analysis.

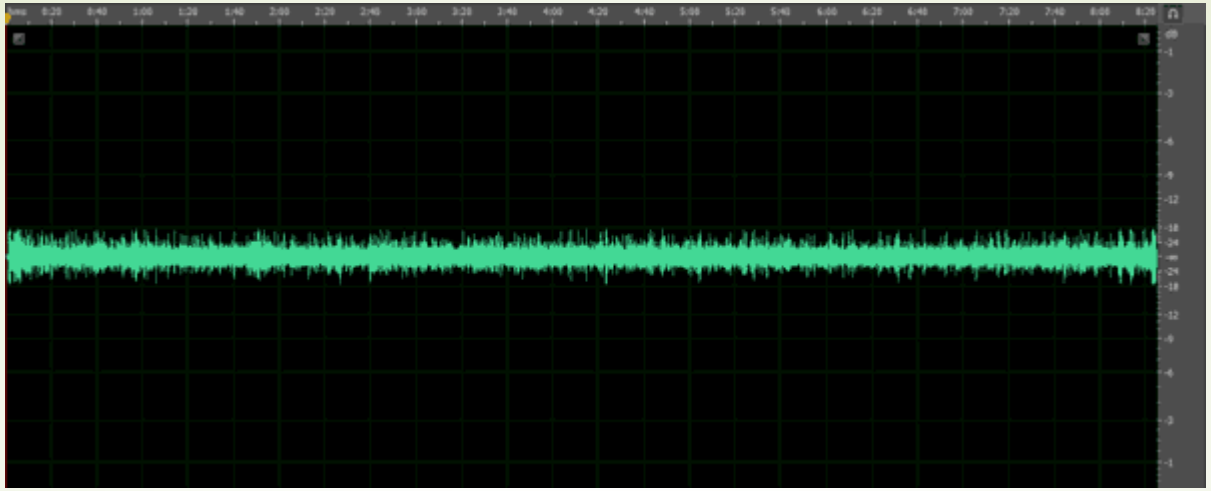
Techniques can include:

- ***Sharpening*** – Makes edges of images in the recording become more clear and distinct.
- ***Video stabilization*** – Reduces the amount of movement in the video, producing the smoothest possible playback.
- ***Masking*** – Covers the face or areas of the video that may protect a witness, victim or law enforcement officer.
- ***Interlacing*** – In an analog system, interlaced scanning is used to record images (a technique **of combining two television fields** in order to produce a full frame of video). A process called **de-interlacing** may be used to retrieve the information in both fields of video.
- ***Demultiplexing*** – Allows for isolation of each camera. In CCTV systems, a device called a multiplexer is used to combine **multiple video signals into a single signal or separate a combined signal**.

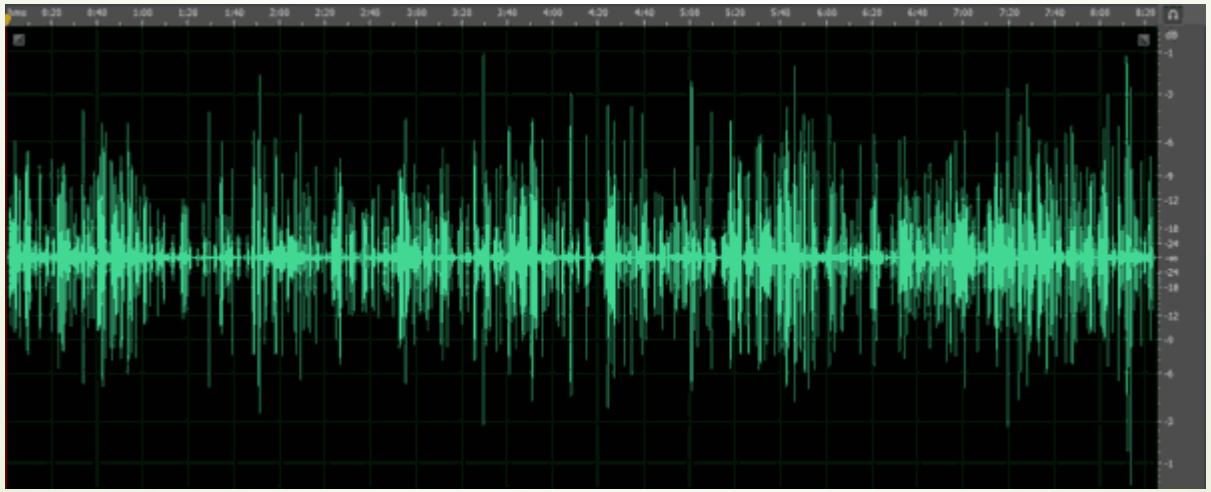
Audio Enhancement Techniques

For audio recordings, a variety of filters can be applied to enhance the material, bringing out specific aspects or events contained in the recording.

- ***Frequency Equalization*** – Highly precise equalizers can be used to boost or cut specific bands of frequencies. To help make speech more intelligible, the frequency band containing **most speech content, 200Hz–5000Hz, can be amplified or isolated**. If amplification is applied to a frequency range, other information residing in this frequency range will be boosted as well.
- If **noise resides in this same range**, this noise will also be increased, limiting the ability to clarify voices.Loud background noises may be analyzed by a **spectrum analyzer** and the corresponding frequencies reduced so that these noises are less noticeable.
- ***Compression*** – Faint sounds in the recording can be boosted by compressing or leveling the signal so that the dynamic range of the material is reduced, making soft sounds more apparent.



Waveform of a recording made at a low volume with significantly loud ambient noise that is masking the speech content of the recording.



The same recording after enhancement. The noise is attenuated and the volume of the speech is increased.

What information does the analysis report contain?

- The nature of the request
- The results that were obtained
- Identification of the format/type of audio/video
- The type of processing that was performed
- Dates and time stamps that were visible
- Description of the subjects, if visible
- Description of the final product provided (e.g. enhanced video prints or video recording)
- Limitations in the examination

Limitations:

- Resolution of the original recorded image
- Low light conditions for video recording
- Noisy environment for audio recording
- Damaged equipments

Companies:

- Audio Video Forensics
- CFS(Creative Forensics)
- CYFOR

Thank You